Percentage increase and decrease

1 Steph is going to buy a hat that is marked as 25% off. The original price was $\$36$.

a What is the value of the discount in dollars?

b What is the price that Steph will pay for the hat?

2 Duncan is going to buy a hat that is marked as 15% off. The original price was $\$30$.

a What is the value of the discount in dollars?

b What is the price that Duncan will pay for the hat?

3 Christa is going to buy an eraser from a stationery store that is having a storewide 10% off sale. The original price of the eraser was $\$6$.

a What is the value of the discount in dollars?

b What is the price that Christa will pay for the eraser?

4 A movie ticket usually costs $\$21$ but staff are given a 40% discount.

a How much less do staff pay for a ticket?

b What price do staff pay for a ticket?

5 A music shop advertises 12% off usual prices on all CDs and DVDs. A CD is usually priced at $\$16$.

a Find the amount of the discount on this CD.

b Calculate the discounted price of the CD.

6 Brad is going to purchase some sports gear on layby. This involves paying some money as a deposit and paying the remainder later. The amount of the layby is $\$180$.

a If the amount the store needs as a deposit is 20%, calculate the dollar amount of the deposit.

b Calculate the remaining balance on the layby.

7 Caitlin received a store discount of 40% on a new bed originally priced at $\$670$.

a What percentage of the price did she pay?

b Calculate the price of the bed after store discount was applied.

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8 A store is offering a 20% discount on all goods. Ivan purchases a fridge with an original price of $\$640$ and a stove with an original price of $\$840$.

a Without a discount, how much would Ivan have to pay altogether?

b Calculate the total discount amount that Ivan receives.

c Find the amount Ivan pays for the goods after discount.

9 A tennis racket marked at a price of $\$90$ is advertised to be selling at 45% off the marked price. Calculate the discounted price.

10 A bag of rice weighs $110\ \text{kg}$. If the weight of the bag decreases by 40%, find the new weight of the bag.

11 A bag of potatoes weighs $50\ \text{kg}$. If the weight of the bag increases by 35%, find the new weight of the bag.

12 A basket of goods was valued at $\$43.70$ in January 2020. If the inflation rate for the year was 8%, find the expected cost of the basket of goods in January 2021.

13 Maria bought a car for $\$5400$ and sold it two years later, making a loss of 15%. How much did she sell it for?

14 A sales assistant is paid a commission of 15% on her weekly sales. Find her commission for a week in which she sells products to the value of $\$2000$.

15 The value of my car depreciated 5% this year. If its value at the start of the year was $\$4000$:

a By how much did it depreciate?

b Calculate the current value of the car.

16 Two stores advertised the same range of books:

- Honest Anne offered 20% off the publisher's recommended price.
- Fair Freddie offered up to 50% off the publisher's recommended price.

a Fiona bought books from Honest Anne's with a total marked price of $\$130$, prior to discount. Find the value of the discount she will receive.

b How much did Fiona pay for the books at Honest Anne's Shop?

c At Fair Freddie's, Donna bought the same books for the same price and was given 10% off the first $\$100$ worth of purchase and 50% off the remainder of the purchase. Calculate the total discount Donna received from Fair Freddie's shop.

d How much did Donna pay for the books at Fair Freddie's Shop?

e Which bookshop offer is the best for this purchase?

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17 Identify which of the following offers gives the greater discount:

a

- Offer A: successive discounts of 6% and 11%
- Offer B: a single discount of 13%

b

- Offer A: a single discount of 16%
- Offer B: successive discounts of 5% and 11%

18 A dishwasher initially selling for $\$866$ is discounted by 12%.

a Calculate the discounted price.

b Express the initial price as a percentage of the discounted price, correct to two decimal places.

c Hence state the percentage increase needed to restore the discounted price back to the original price.

19 A holiday resort in Tasmania reduced its overnight rate from $\$232$ to $\$120$.

a Find the amount that Beth would save if she is to take advantage of the sale.

b Find the percentage discount, correct to two decimal places.

20 A bike regularly sells for $\$300$ but is now on sale for $\$255$.

a By how much has the price of the bike been decreased?

b Calculate the percentage discount on the bike.

21 Every item in store is sold at 8% more than its cost. If a product was bought from the manufacturer for $\$360$, calculate:

a The sale price.

b The profit in dollars.

22 A game retailer sells new games at a markup of 33% above the cost price and old games at a markup of 18%. Calculate the selling price of:

a The newly released Super Zora 5 game which has a cost price of $\$90$.

b The old Super Zora 3 game which has a cost price of $\$31$.

23 At the end of the financial year, a television originally advertised at $\$3800$ is discounted by 5%. Valentina negotiates a further discount for paying cash, which brings the price down to $\$3285.10$. Calculate the further percentage discount Valentina received for paying cash.

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24 Luke buys a used bike for $\$385$ and resells it for $\$560$ without spending any money on it.

- **a** How much profit did Luke make on the bike?
- **b** Calculate Luke's percentage profit, correct to two decimal places.

25 Sally purchased a dress for $\$20$ and auctioned it online for $\$41$. Calculate:

- **a** The profit as a percentage of the cost, correct to two decimal places.
- **b** The profit as a percentage of the revenue, correct to two decimal places.

26 Every item in store is sold at 9% more than its cost. If a product is sold for $\$110$, calculate:

- **a** The cost price.
- **b** The profit.

Calculate original values

27 A woman is paid a commission of 9% on her weekly sales. If her total commission for the week was $\$541$, calculate:

- **a** The value of 1% of her sales.
- **b** The total value of her sales for that week to the nearest dollar.

28 A store is having a 13% off sale. Calculate the original price of a discounted item that is on sale for $\$3723.75$.

29 A store is having a 15% off sale. Calculate the original price of a discounted item that is on sale for $\$495$.

30 An investor sold his portfolio at 8% profit for a price of $\$2000$. Calculate the original value of his portfolio.

31 In a fire sale, a store sold goods at 41% below cost price. If $\$13\,900$ was earned from the sale, calculate:

- **a** The original cost of goods sold.
- **b** The loss on the sale.

32 A manufacturer is selling its old factory machinery for $\$34\,320$, which represents a loss of 12%. At what price did the manufacturer purchase the machinery?

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Goods and services tax

33 Calculate the 10% GST applicable on the following original prices:

a $\$80$ **b** $\$39.50$ **c** $\$35$ **d** $\$77.80$

34 Calculate the sales price, including GST, for an item with an original price of $\$20$.

35 John received a bill of $\$68$ for consultation and $\$10$ for herbal supplements from a medical professional. If GST is only applied to the consultation fee, what is the total bill that John must pay?

36 Calculate the GST on an item with a sales price (including GST) of:

a $\$40$ **b** $\$83.90$

37 The sales price of an item, including GST, is $\$80$. Calculate the price of the item without GST.

38 Fruits and vegetables do not incur GST of 10%. If the total shopping bill is $\$109$, and fruits and vegetables amount to $\$62$, what is the cost of the other items prior to adding GST?

39 Calculate the amount of GST paid on the receipt shown.

Note: The prices shown already include GST.

Humble Hut
12 Miller Rd, Perth
WA, 6000

BILL FOR TABLE 306
ABN: 90 082 402 695
Printed by: Michael
Started by: Freedham
Printed from: POS9 - VP Dining
Date: 06/09/19 18:53

Item	Qty	Price
*Tofu scramble	1	$11.50
*Buckwheat pancakes	1	$9.00
*Acai bowl	1	$8.00
TOTAL INCLUDES GST		$28.50
*GST item		

We do not split bills

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Shares, dividends and yields

40 The following table highlights the shares Dave owns in a number of companies:

	Share Price (dollars)	Number of Shares Owned	Value of Shares (dollars)	Dividend Yield
Straits Airways	1.68	800	A	2%
Outback Mining Co.	20.45	200	B	5.4%
Bargain Supermarkets	14.34	300	C	4.8%
Bank of Oz	11.22	150	D	3%
Insurance Corp.	7.60	600	E	1.25%

a Calculate the value of:

i A **ii** B **iii** C **iv** D

v E

b Calculate the total value of shares that Dave owns.

c Calculate the dividend per share earned on shares in the company 'Straits Airways'.

d Calculate the total dividend earned on Dave's shares in the company 'Bargain Supermarkets'. Round your answer to the nearest whole dollar.

41 The annual earnings per share is $\$1.25$ and the market price of the share is $\$12.25$.

Calculate the price-to-earnings ratio.

42 The market value of shares in Florist Gump is $\$18.55$ per share. The P/E ratio is 10.6.

Calculate the annual earnings per share.

43 The P/E ratio of shares in Vinyl Resting Place is 9.4. The annual earnings per share is $\$1.75$.

Calculate the price of each share.

44 Calculate the dividend per share when a company's net profit of $\$373\,520$ is to be distributed evenly among its $128\,800$ shares.

45 Calculate the dividend yield, correct to one decimal place, when a firm with a share price of $\$38.20$ pays a dividend of $\$1.38$ per share.

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46 Calculate the dividend per share when a company's share price is $\$34.00$ and its dividend yield is 5.3%.

47 Calculate the share price of a company that pays a dividend of $\$3.00$ per share and has a dividend yield of 2.8%.

48 Consider the following companies share and dividend information:

- Company A has a share value of $\$2.50$ and paid a dividend of $\$0.19$ per share.
- Company B has a share value of $\$8.40$ and paid a dividend of $\$0.63$ per share.
- Company C has a share value of $\$16.25$ and paid a dividend of $\$1.43$ per share.

a Calculate the dividend yield for Company A.

b Calculate the dividend yield for Company B.

c Calculate the dividend yield for Company C.

d Which company had the highest dividend yield?

49 Sean buys 6000 shares in Company A at $\$6.25$ each. The company pays a dividend of $55c$ per share. Sean also buys 3000 shares in Company B at $\$7.50$ per share, which pays a dividend of $\$0.57$ per share.

a Calculate the dividend yield for company A.

b Calculate the dividend yield for company B.

c What is the overall dividend yield for Sean's investment?

50 Amelia buys shares in a company at $\$7.25$ per share. The company pays a dividend in July of $27.5c$ and a dividend in February of $27.6c$. What is the dividend yield for the whole financial year?

51 A company has a net profit of $\$13.2$ million that it will fully distribute to the owners of its 82.5 million shares.

a Calculate the dividend per share that will be paid to shareholders.

b Calculate the dividend yield if the share price is $\$26.37$. Round your answer to one decimal place.

c In order to raise funds, the company issues another 2.2 million shares. Given that the company expects its profits to increase by 1% in the next year and it intends to distribute all of its net profit to shareholders, what will its dividend yield be if its share price remains the same? Round your answer to one decimal place.

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52 Mastaneh has a share portfolio as detailed below. Assume that all earnings are distributed as dividends.

Company	Number of shares	Price per share	Dividend per share	Dividend yield
Sham Rock Diamonds	300	\$1.76	y	12.5%
The Codfather	700	\$1.25	\$0.08	x
Curl Up and Fry	900	\$1.25	\$0.04	3.2%

a Calculate the total value of this share portfolio.

b Calculate the value of x. Express your answer as a percentage.

c Calculate the value of y.

d Calculate the price-to-earnings ratio of Sham Rock Diamonds.

e Calculate the price-to-earnings ratio of The Codfather.

f Calculate the price-to-earnings ratio of Curl Up and Fry.

g Based on the price-to-earnings ratio, which company is the best investment?

53 Freya paid \$7800 for 300 shares that have a dividend yield of 2.5%. The company returned 52% of its profits to shareholders as a dividend.

a Calculate the price per share.

b Calculate the dividend per share.

c Calculate the total earnings per share.

d Calculate the price-to-earnings ratio.

54 The table below summarises information about four companies:

Company	Share price	Earnings per share	P/E	Dividend per share	Dividend yield
True Blue Holdings	\$10.25	\$1.25	a	\$0.287	2.8%
Sew What	\$12.75	b	3.4	\$0.612	4.8%
Juan In A Million	\$17.25	\$3.75	4.6	c	2.8%
Wok On Water	\$22.75	\$8.75	2.6	\$1.547	d

Calculate the value of:

a a **b** b **c** c **d** d

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